

Model Card: Multinomial Naive Bayes

Audit: bias & explainability

Audit 1 of 4 | Model: Multinomial Naive Bayes | **Config:** MultinomialNB() with default smoothing (alpha = 1).

Features: TF-IDF, 1-2-grams, top 3,000 | **Threshold:** 0.50 | **Data:** UCI SMS Spam Collection, 403 duplicates removed; 4,135 train / 1,034 test (stratified 80/20, seed 42); spam 12.6%.

Intended use: educational demo / triage of English SMS. **Not for** automatic blocking without human review.

Performance and subgroup audit

Overall (held-out test): accuracy 0.975, precision 1.000, recall 0.802, F1 0.890, ROC-AUC 0.986, Brier 0.022. **0** false positives, **26** false negatives (of 131 spam).

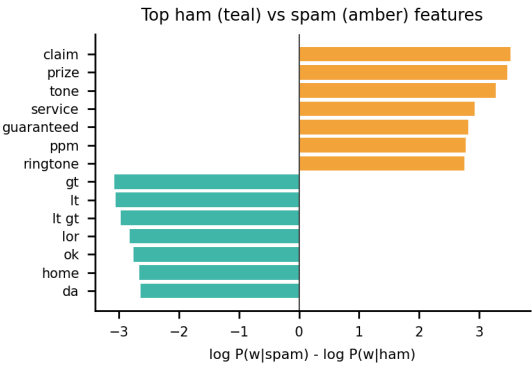
Subgroup	n	Spam	Precision	Recall	F1	FPR	FP	FN
All test messages	1034	131	1.00	0.80	0.89	0.000	0	26
Length: short (<=42 ch)	353	2	–	0.00	0.00	0.000	0	2
Length: medium (43-91 ch)	346	12	1.00	0.50	0.67	0.000	0	6
Length: long (>91 ch)	335	117	1.00	0.85	0.92	0.000	0	18
Style: URL / 5+ digit number	107	107	1.00	0.86	0.92	–	0	15
Style: currency or digits	252	125	1.00	0.82	0.90	0.000	0	23
Style: ALL-CAPS heavy (>30%)	44	19	1.00	0.79	0.88	0.000	0	4
Style: casual / slang	340	37	1.00	0.70	0.83	0.000	0	11
Style: plain (none of above)	539	6	1.00	0.50	0.67	0.000	0	3

Groups overlap (a message can be in several style groups). FPR = false-positive rate; '–' = undefined. Red = recall < 0.60 with ≥5 spam, or precision < 0.60 with ≥5 false positives.

Subgroup findings. The model is extremely cautious: **zero false positives** in every group (precision 1.00), so ordinary chat is never wrongly flagged. The price is recall. Casual / slang spam is caught only **70%** of the time (11 of 37 missed) versus 95%+ for long and URL/number messages in the stronger models, and medium-length spam only 50% (6 of 12). *Caution:* short (2 spam) and plain (6 spam) groups are too small to trust.

Explainability and failure analysis

Explainability (log-probability ratio, the model's own parameters). Spam is identified by a narrow, era-specific promo vocabulary: *claim, prize, tone, service, guaranteed, ppm, ringtone, uk*. Ham is identified by chat words and the HTML-entity artefacts *gt / lt* (from <>). **Failure analysis:** 26 spam were missed with a median spam probability of only 0.14, so they do not even come close to the threshold. Lowering the threshold barely helps: at 0.35 recall is only 0.84 (+6 FP).



Limitations & intended use

Limitations. (1) **Vocabulary lock-in:** a spam message that avoids the learned promo words looks like ham, and the model fails silently (no false alarms, just misses). (2) Independence assumption makes probabilities poorly spread, so thresholds are an ineffective tuning knob. (3) Misses spam of the slang, adult-chat and missed-call type. **Use when:** a false alarm is much worse than a missed spam message.

Shared caveat (see page 5): a preprocessing issue removes URL / currency / number cues for all four models.

Model Card: Logistic Regression

Audit: bias & explainability

Audit 2 of 4 | Model: Logistic Regression | **Config:** LogisticRegression(max_iter=1000, class_weight='balanced').
Features: TF-IDF, 1-2-grams, top 3,000 | **Threshold:** 0.50 | **Data:** UCI SMS Spam Collection, 403 duplicates removed; 4,135 train / 1,034 test (stratified 80/20, seed 42); spam 12.6%.
Intended use: educational demo / triage of English SMS. **Not for** automatic blocking without human review.

Performance and subgroup audit

Overall (held-out test): accuracy 0.971, precision 0.863, recall 0.916, F1 0.889, ROC-AUC 0.991, Brier 0.035. **19** false positives, **11** false negatives (of 131 spam).

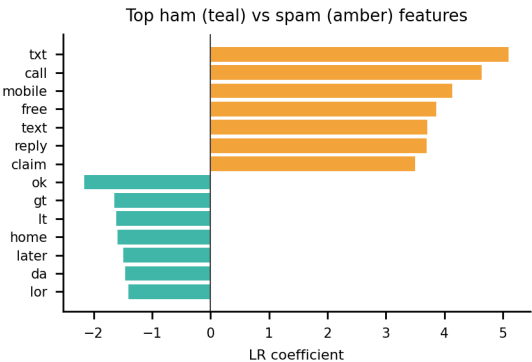
Subgroup	n	Spam	Precision	Recall	F1	FPR	FP	FN
All test messages	1034	131	0.86	0.92	0.89	0.021	19	11
Length: short (<=42 ch)	353	2	0.08	0.50	0.14	0.031	11	1
Length: medium (43-91 ch)	346	12	0.53	0.67	0.59	0.021	7	4
Length: long (>91 ch)	335	117	0.99	0.95	0.97	0.005	1	6
Style: URL / 5+ digit number	107	107	1.00	0.96	0.98	—	0	4
Style: currency or digits	252	125	0.99	0.93	0.96	0.008	1	9
Style: ALL-CAPS heavy (>30%)	44	19	1.00	0.95	0.97	0.000	0	1
Style: casual / slang	340	37	0.89	0.89	0.89	0.013	4	4
Style: plain (none of above)	539	6	0.21	0.67	0.32	0.028	15	2

Groups overlap (a message can be in several style groups). FPR = false-positive rate; '—' = undefined. Red = recall < 0.60 with ≥5 spam, or precision < 0.60 with ≥5 false positives.

Subgroup findings. Best recall on structured spam (URL/number 0.96, long 0.95) but it **over-flags short, plain conversation**: 11 false positives among short messages (precision **0.08**) and 15 among plain messages (precision **0.21**). People who write brief, everyday requests ("call me", "are you free") are the group most unfairly treated. Casual/slang text gets 4 false alarms (FPR 0.01). *Caution:* short and plain groups hold only 2 and 6 spam messages.

Explainability and failure analysis

Explainability (coefficients). Spam cues: *txt, call, mobile, free, text, reply, claim, prize*. Ham cues: *ok, home, later, da, lor*. **Failure analysis:** Most of the 19 false positives are normal messages containing a spam word: "Call me when u're done..." (0.55), "Are you free now?can i call now?" (0.83), "Where are you call me." (0.95), "I'm at work. Please call" (0.80). Everyday words that spammers also use get treated as spam evidence. The 11 misses sit near the boundary (median p 0.43).



Limitations & intended use

Limitations. (1) **Word-level bias:** *call, text, free, reply* carry large positive weights because 2000s spam used them, so benign requests to call or text are penalised. (2) *class_weight='balanced'* trades precision for recall: raising recall to 0.97 at threshold 0.4 costs 29 false positives. (3) No interaction between words ("call me" vs "call now"). **Use when:** missing spam is costly and a human reviews flagged messages.
Shared caveat (see page 5): a preprocessing issue removes URL / currency / number cues for all four models.

Model Card: Linear SVM

Audit: bias & explainability

Audit 3 of 4 | Model: Linear SVM | **Config:** CalibratedClassifierCV(LinearSVC(class_weight='balanced')), Platt-scaled probabilities.
Features: TF-IDF, 1-2-grams, top 3,000 | **Threshold:** 0.50 | **Data:** UCI SMS Spam Collection, 403 duplicates removed; 4,135 train / 1,034 test (stratified 80/20, seed 42); spam 12.6%.
Intended use: educational demo / triage of English SMS. **Not for** automatic blocking without human review.

Performance and subgroup audit

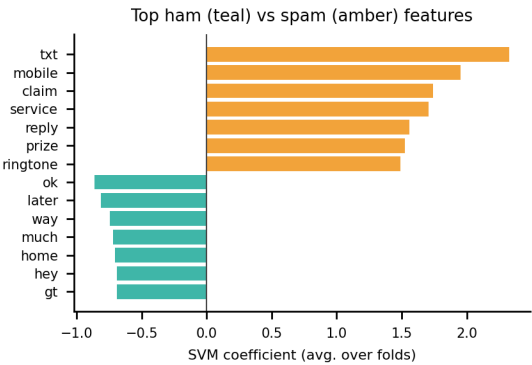
Overall (held-out test): accuracy 0.982, precision 0.938, recall 0.916, F1 0.927, ROC-AUC 0.994, Brier 0.015. **8** false positives, **11** false negatives (of 131 spam).

Subgroup	n	Spam	Precision	Recall	F1	FPR	FP	FN
All test messages	1034	131	0.94	0.92	0.93	0.009	8	11
Length: short (<=42 ch)	353	2	0.20	0.50	0.29	0.011	4	1
Length: medium (43-91 ch)	346	12	0.73	0.67	0.70	0.009	3	4
Length: long (>91 ch)	335	117	0.99	0.95	0.97	0.005	1	6
Style: URL / 5+ digit number	107	107	1.00	0.96	0.98	—	0	4
Style: currency or digits	252	125	0.99	0.93	0.96	0.008	1	9
Style: ALL-CAPS heavy (>30%)	44	19	1.00	0.95	0.97	0.000	0	1
Style: casual / slang	340	37	0.94	0.89	0.92	0.007	2	4
Style: plain (none of above)	539	6	0.40	0.67	0.50	0.011	6	2

Groups overlap (a message can be in several style groups). FPR = false-positive rate; '—' = undefined. Red = recall < 0.60 with ≥5 spam, or precision < 0.60 with ≥5 false positives.
Subgroup findings. Strongest overall balance. Long, URL/number, currency and ALL-CAPS groups all have recall of 0.93-0.95 and precision of 0.99-1.00. Weak spots mirror LR but are milder: short messages give 4 false positives (precision **0.20**), plain messages 6 (precision **0.40**), and medium messages only 0.73 precision / 0.67 recall. Casual/slang text does well (F1 0.92). *Caution:* short and plain have only 2 and 6 spam messages.

Explainability and failure analysis

Explainability (coefficients averaged over calibration folds). Spam: *txt, mobile, claim, service, reply, prize, ringtone* and, notably, *arsenal*. Ham: *ok, later, way, much, home, hey*. *Arsenal* shows the model memorises topic-specific promotions (football-ticket competitions). **Failure analysis:** 8 false positives, e.g. "K k:) sms chat with me." scored 0.97, a confident mistake caused by *sms/chat*; 11 misses (median p 0.16) are largely spam without the model's strong cue words.



Limitations & intended use

Limitations. (1) **Confident errors:** the worst false positive scores 0.97, so probability alone does not signal when to doubt it. (2) **Topic memorisation** (e.g. *arsenal*) will not transfer to new campaigns. (3) Threshold changes help little (0.25 gives recall 0.95 for 23 FP). **Use when:** a good default is needed. Best F1 (0.93), accuracy (0.98) and calibration (Brier 0.015).
Shared caveat (see page 5): a preprocessing issue removes URL / currency / number cues for all four models.

Model Card: Random Forest

Audit: bias & explainability

Audit 4 of 4 | Model: Random Forest | **Config:** RandomForestClassifier(n_estimators=200, class_weight='balanced').
Features: TF-IDF, 1-2-grams, top 3,000 | **Threshold:** 0.50 | **Data:** UCI SMS Spam Collection, 403 duplicates removed; 4,135 train / 1,034 test (stratified 80/20, seed 42); spam 12.6%.
Intended use: educational demo / triage of English SMS. **Not for** automatic blocking without human review.

Performance and subgroup audit

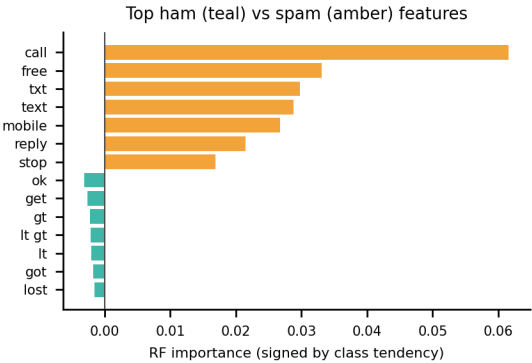
Overall (held-out test): accuracy 0.977, precision 0.973, recall 0.840, F1 0.902, ROC-AUC 0.994, Brier 0.021. **3** false positives, **21** false negatives (of 131 spam).

Subgroup	n	Spam	Precision	Recall	F1	FPR	FP	FN
All test messages	1034	131	0.97	0.84	0.90	0.003	3	21
Length: short (<=42 ch)	353	2	0.00	0.00	0.00	0.006	2	2
Length: medium (43-91 ch)	346	12	1.00	0.33	0.50	0.000	0	8
Length: long (>91 ch)	335	117	0.99	0.91	0.95	0.005	1	11
Style: URL / 5+ digit number	107	107	1.00	0.90	0.95	—	0	11
Style: currency or digits	252	125	1.00	0.86	0.92	0.000	0	18
Style: ALL-CAPS heavy (>30%)	44	19	1.00	0.84	0.91	0.000	0	3
Style: casual / slang	340	37	1.00	0.84	0.91	0.000	0	6
Style: plain (none of above)	539	6	0.50	0.50	0.50	0.006	3	3

Groups overlap (a message can be in several style groups). FPR = false-positive rate; '—' = undefined. Red = recall < 0.60 with ≥5 spam, or precision < 0.60 with ≥5 false positives.
Subgroup findings. Recall is uneven. Long messages 0.91, but medium messages only **0.33** (8 of 12 spam missed). Spam with a URL or 5+ digit number 0.90, so the model leans on these cues. Plain, short messages are the only place false positives appear (3 of 3). Casual/slang and ALL-CAPS text do **not** raise false alarms (FPR 0.000). *Caution:* short and medium groups hold only 2 and 12 spam.

Explainability and failure analysis

Explainability (feature importance, direction added from class means). Spam: *call, free, txt, text, mobile, reply, stop, claim*. Ham: *ok, get, got, home*. Note: RF importance has no sign, so direction is inferred from whether a word is more common in spam training messages. **Failure analysis:** the 3 false positives ("I'm at home. Please call", "I'm at work. Please call", a thank-you note with "service") come from *call/service*. Many of the 21 misses are adult-chat and premium-rate lines at p 0.2-0.47, just under 0.5.



Limitations & intended use

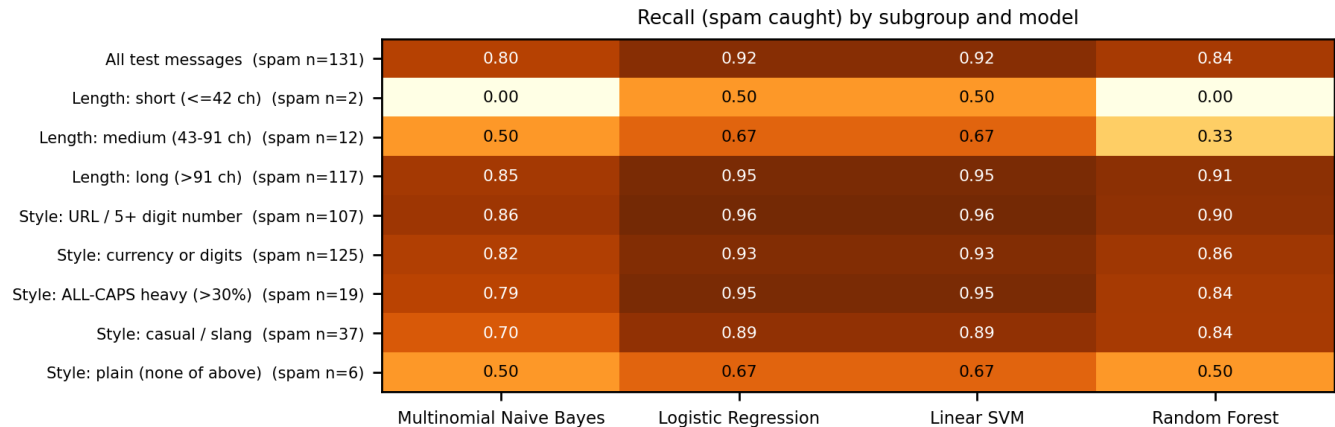
Limitations. (1) Under-detection: 21 of 131 spam missed (recall 0.84), most scoring just below 0.5. Threshold 0.35 lifts recall to 0.92 at 7 false positives. **(2)** Trees on 3,000 sparse TF-IDF features rely on a few strong words, and importances are a coarse explanation. **(3)** Averaged votes give coarse probabilities. **Use when:** false alarms are costly but some missed spam is acceptable.
Shared caveat (see page 5): a preprocessing issue removes URL / currency / number cues for all four models.

Final Audit: All Models Combined

Audit 5 of 5: cross-model comparison, shared findings and recommendation

Model comparison (held-out test set, threshold 0.50; best F1 highlighted)

Model	Acc	Prec	Recall	F1	ROC-AUC	Brier	FP	FN	Worst subgroup issue
Multinomial Naive Bayes	0.975	1.000	0.802	0.890	0.986	0.022	0	26	Slang spam recall 0.70
Logistic Regression	0.971	0.863	0.916	0.889	0.991	0.035	19	11	Short/plain precision 0.08/0.21
Linear SVM	0.982	0.938	0.916	0.927	0.994	0.015	8	11	Short/plain precision 0.20/0.40
Random Forest	0.977	0.973	0.840	0.902	0.994	0.021	3	21	Medium recall 0.33
Soft-vote ensemble (mean of 4)	–	0.967	0.885	0.924	0.995	–	4	15	Not better than SVM on F1



Bias pattern across models. Errors are not spread evenly. **False positives concentrate in short, plain, conversational messages** (the 'plain' group has only 6 spam but produces most false alarms in LR, SVM and RF). **Missed spam concentrates in medium-length, slang and adult/premium-rate messages.** Of 131 spam messages, **11** (8%) are missed by **all four** models and 30 by at least one; 20 ham messages are flagged by at least one model, but **none by all four**. So most errors are model-specific (the models disagree), but a hard core of spam is invisible to all of them.

Explainability across models

All four models rely on the same cue words: *txt, call, text, mobile, free, reply, claim, prize* for spam and *ok, home, later* for ham. That agreement is reassuring (they learned genuine spam language) but it is also the shared weakness: **benign messages containing 'call', 'text' or 'free' are flagged**, and spam without those words is missed. SVM additionally memorised a topic word (*arsenal*), a sign of overfitting to specific campaigns.

Key finding: preprocessing removes the URL / currency / number cues

In `clean_text`, the text is lowercased first, then URL, CURRENCY and NUM are inserted in capitals, and the next line `[^a-z\s]` deletes every capital letter. The placeholders never reach the model (e.g. 'Call 09099726395 now at www.x.com for £100' becomes just 'call'). This is likely why the shared misses are phone-number-driven lines. As a test only (your app is unchanged), using lowercase placeholders (`urltoken, currencytoken, numtoken`) gives:

Model	Prec	Recall	F1	ROC-AUC	FP	FN	F1 before
Multinomial Naive Bayes	1.000	0.840	0.913	0.992	0	21	0.890
Logistic Regression	0.947	0.947	0.947	0.997	7	7	0.889
Linear SVM	0.992	0.947	0.969	0.998	1	7	0.927
Random Forest	1.000	0.916	0.956	0.998	0	11	0.902

Recommendation. Fix the placeholder issue first; it improves every model, most for Linear SVM (F1 0.927 to 0.969, 1 false positive). Before and after the fix, **Linear SVM is the best single model** (best F1 and Brier score); the soft-vote ensemble does not beat it on F1. Choose NB if false alarms are unacceptable, LR if missed spam is the bigger risk. Always keep a human in the loop.

Limitations of this audit. (1) The test set has only 131 spam messages (short: 2, plain: 6, medium: 12), so subgroup numbers are noisy and there are no confidence intervals. (2) Dated, UK-centric, English-only data; modern delivery and bank-link scams are not represented. (3) Subgroups are heuristic regex rules and overlap. (4) One random split. (5) Explanations are model-specific proxies (RF has no signed weights), not SHAP/LIME.